

- 6 (a) e.m.f.: energy converted from chemical / other forms to electrical
per unit charge B1
p.d.: energy converted from electrical to other forms per unit charge B1 [2]
- (b) (i) the p.d. across the lamp is less than 12 V
or there are lost volts / power / energy in the battery / internal resistance B1 [1]
- (ii) $R = V^2 / P$ (or $V = RI$ and $P = VI$) C1
= 144 / 48
= 3.0 Ω A1 [2]

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(iii) $I = E/(R_T + r)$
 $= 12/2.0$
 $= 6.0 \text{ A}$

C1

A1 [2]

(iv) power of each lamp = $I^2 R$
 $= (3.0)^2 \times 3.0$
 $= 27 \text{ W}$

C1

A1 [2]

(c) less resistance (in circuit)/more current
more lost volts/less p.d. across battery

M1

A1 [2]